Mecánica Analítica

Grupo 8222– 2026–1

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Tarea 2

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Problema 1

El desplazamiento de un objeto con masa m sujeta a un resorte bajo la acción de la fricción de Coulomb satisface la siguiente ecuación,

$$\ddot{x} + \Omega^2 x = \begin{cases} -F_0 & \dot{x} > 0 \\ F_0 & \dot{x} < 0 \end{cases} \tag{1.1}$$

con Ω y F_0 constantes positivas. Si $|x| > F_0/\Omega^2$ cuando $\dot{x} = 0$, el objeto continúa en movimiento mientras que si $|x| \leq F_0/\Omega^2$ cuando $\dot{x} = 0$, entonces el objeto se detiene. Inicialmente el cuerpo empieza en reposo con $x = 9F_0/2\Omega^2$. Encuentra donde se detiene la masa y por cuanto tiempo estuvo en movimiento.

Solución

Problema 2

Un oscilador forzado satisface la siguiente ecuación

$$\ddot{x} + \Omega^2 x = F_0 \cos\Big[\Omega(1+\epsilon)t\Big] \tag{2.1}$$

con ϵ una constante positiva. Muestra que la solución que satisface las condiciones iniciales $x = 0$ y $\dot{x} = 0$ cuando $t = 0$ es

$$x = \frac{F_0}{\epsilon\Big(1+\frac{1}{2}\Big)\Omega^2} \sin\Big[\tfrac{1}{2}\epsilon\Omega t\Big] \sin\Big[\Omega(1+\epsilon)t\Big]. \tag{2.2}$$

Grafica $x(t)$ cuando ϵ es pequeña.

Solución

Problema 3

Un oscilador sobreamortiguado satisface la ecuación

$$\ddot{x} + 10\dot{x} + 16x = 0. \tag{3.1}$$

En $t = 0$ la partícula está en $x = 1$ y sale con dirección hacia el origen con velocidad u. Encuentra la trayectoria de dicha partícula. Muestra además que la partícula llegará al origen de coordenadas si

$$\frac{u-2}{u-8} = \mathrm{e}^{6t}. \tag{3.2}$$

¿Qué tan grande tiene que ser u para que la partícula se pase del origen de coordenadas?

Solución

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions. (Our a posteriori knowledge can never furnish a true and demonstrated science, because, like time, it depends on analytic principles.) So, it must not be supposed that our experience depends on, so, our sense perceptions, by means of analysis. Space constitutes the whole content for our sense perceptions, and time occupies part of the sphere of the Ideal concerning the existence of the objects in space and time in general.

Problema 4

Una partícula de masa m, se mueve sobre el eje negativo x hacia el origen de coordenadas con velocidad constante u. Cuando la partícula llega al origen siente una fuerza $F = -kx^2$, con k una constante positiva. Calcula la distancia máxima que avanza la partícula sobre el positivo x.

Solución